

RESEARCH LETTER – Food Microbiology

Comparative analysis of Methicillin-Resistant *Staphylococcus aureus* (MRSA) and Borderline Oxacillin Resistant *Staphylococcus aureus* (BORSA) in community and food of animal origin

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One sentence summary: Current study provides the information on the statistical relationship between the genetic determinants of *S. aureus* with respect to sample type, and additionally the correlation that exists between the *pvl* and MRSA, *pvl* and cotrimoxazole resistance, vancomycin MIC and MRSA/Methicillin-Susceptible *S. aureus* (MSSA).

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ABSTRACT

The aim of this study was to illustrate the relative pervasiveness of Borderline Oxacillin Resistant *Staphylococcus aureus* (BORSA) and Methicillin-Resistant *Staphylococcus aureus* (MRSA) in community and food of animal origin and their relationship with other genetic determinants. *Staphylococcus aureus* isolates were subjected to E-test using the antibiotics: oxacillin, ceftriaxone, cotrimoxazole, vancomycin, genotypic tests for the genes *mecA*, *vanA*, *blaZ*, *pvl* gene and *SCCmec* typing. The prevalence of *S. aureus* (MRSA) in the food of animal origin and community settings was 21% (1.8% MRSA) and 21.9% (7.4% MRSA), respectively. *SCCmec* type V was prevalent among the food of animal origin, while *SCCmec* type IVa among the community isolates. The likelihood of MRSA presence among community isolates was three times more than in isolates from chicken and milk samples. Likewise, the likelihood of detecting *pvl* positive MRSA (*pvl*⁺MRSA) isolates was 4-fold higher in the community setting than in the food of animal origin. The *mecA* negative BORSA (*mecA*⁻BORSA) was a frequently observed phenotype among *S. aureus* isolates. Also, co-detection of *pvl* and cotrimoxazol resistance was reported in this study although there was no noteworthy correlation of cotrimoxazol resistance with the type of sample. Isolates from milk and community settings exhibit higher minimum inhibitory concentration to vancomycin (Vancomycin MIC creep, 2–4 µg/mL).

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Significance: Current study provides the information on the statistical relationship between the genetic determinants of *S. aureus* with respect to sample type, and additionally the correlation that exists between the *pvl* and MRSA, *pvl* and cotrimoxazol resistance, vancomycin MIC and MRSA/Methicillin-Susceptible *S. aureus* (MSSA).

Keywords: antibiotics; BORSA; MRSA; *pvl*; SCCmec typing

INTRODUCTION

Antibiotics are utilized broadly in a sustenance animal production framework, sub-therapeutically for growth promotion and disease prevention, which has brought about the ascent of antibiotic-resistant micro-organisms. Antibiotic-resistant *Staphylococcus aureus* in food animals and foods of animal origin are common because of its commensal association (Haaber, Penadés and Ingmer 2017). Even though it could be found in the animals and environment, humans serve as a major reservoir for *S. aureus* (Moellering 2006). Antibiotic-resistant *S. aureus* especially MRSA has been isolated from retail milk/meat, milk/meat products, meat industry workers, hospital environment and community settings (Sato et al. 2017; Abdulsalam et al. 2018; Titouche et al. 2019).

Borderline oxacillin resistant *S. aureus* (BORSA) is another, relatively frequent phenotype among *S. aureus*. Based on *mecA* gene, BORSA can be classified as *mecA*⁻BORSA (Oxacillin MIC 4–8 µg/mL; Skinner et al. 2008) and *mecA*⁺BORSA (Oxacillin MIC 0.5–8 µg/mL; Huang et al. 2018; Quijada et al. 2019; Fig. 1). The *mecA*⁻BORSA are ceftioxin/ceftriaxone susceptible and do not carry *mecA* or *mecC* gene. These isolates cannot be categorized either genuinely methicillin-resistant or exactly methicillin-susceptible strains. However, they are often misidentified, posing an undeniable epidemiological and therapeutic threat. The accessible proof proposes that *mecA*⁻BORSA represent a frequently neglected problem and their rise in new environments infer that they should be checked and precisely recognized from MSSA and MRSA.

Another virulence factor is Panton-Valentine leukocidin (PVL) that is a cytotoxin and have been repeatedly associated with community-acquired MRSA (CA-MRSA) infections (Darboe et al. 2019). Additionally, PVL presence was associated with the cotrimoxazol resistance (Kraef et al. 2015). Cotrimoxazol prophylaxis has been used to prevent infections in patients infected with the human immunodeficiency virus (HIV). Since cotrimoxazol is a broad-spectrum antibacterial drug so it might have an impact on *S. aureus* colonization in HIV patients (Suthar et al. 2011) thereby, puts HIV patients in danger of skin and soft tissue infections (SSTIs).

This study aimed to determine the occurrence and relationship between the phenotypic and genetic determinants of *S. aureus* with respect to the type of sample under study. Additionally, to identify the correlation that exists between the *pvl* and MRSA, *pvl* and cotrimoxazol resistance, BORSA and ceftriaxone resistance, vancomycin MIC and MRSA/MSSA. In view of SCCmec typing, the accentuation as on the recognition of oxacillin-resistant isolates (i.e. BORSA and MRSA).

MATERIAL AND METHODS

A total of 761 *S. aureus* isolates from the culture collection of the School of Public Health and Zoonoses, GADVASU, Ludhiana, Punjab, India were revived. The isolates belonged to, 432 from meat (chicken, chevon and pork), 126 from raw milk and 203 from community settings (hands, mobile phone, door handle, computer keyboard and mouse and chair's arm-rest). The source of

milk and meat samples was retail markets of different districts of Punjab, community settings isolates were from places around those retail-advertises (Table 1 and Table S2, Supporting Information).

Isolation of *S. aureus* from samples was done as per the methodology explained in previous studies (Kaur et al. 2019; Zehra et al. 2019a). All the revived isolates and the isolates from this study were reconfirmed for *S. aureus* by using HiStaph Identification Kit (HiMedia Labs, Mumbai) for the biochemical characterization of these isolates (Table S1, Supporting Information). ATCC 25923 was used as a reference strain.

Among 761 *S. aureus* isolates, only 277 isolates were selected for this study. The isolates were selected based on following criteria: Non-duplicate; From different regions within a district of Punjab; From different districts of Punjab; Meat samples from retail shops; Raw milk samples from dairy farms/retail shops; Samples from community settings were from places around those retail markets; Biochemical profile of isolates; Collected from August 2014 to July 2017 (Table S2, Supporting Information).

Genomic DNA extraction

The HiPurA bacterial genomic DNA purification kit (HiMedia Lab) was utilized for extraction of the total nucleic acid. The extracted DNA was stored under refrigeration at – 20°C for further analysis.

Molecular identification of *S. aureus*

The *S. aureus* isolates were further confirmed employing molecular detection method, Polymerase Chain Reaction (PCR) using primers targeting conserved sequences like genus specific 16S rDNA and species-specific *fem* gene. The multiplex PCR for the detection of 16S rRNA, *fem* and *mecA* genes was performed as per Zehra et al. 2017.

Antibiotic susceptibility testing of *S. aureus* isolates

The antibiotic susceptibility testing (AST) of *S. aureus* isolates was performed by the E-test (Ezy MIC Strip; HiMedia) to the following antibiotics viz. oxacillin, ceftriaxone, cotrimoxazol and vancomycin according to the manufacturers' recommendation. AST for amoxicillin/clavulanic acid (co-amoxiclav) was performed by standard disc diffusion method according to Clinical and Laboratory Standards Institute (CLSI) guidelines (CLSI 2017). For AST, *S. aureus* ATCC 25923 was used as a reference strain and quality control of the antibiotics was performed as per the manufacturer's guidelines.

Identification of antimicrobial resistance genes, *pvl* gene and SCCmec type

To identify the presence of antibiotic resistance genes, SCCmec type and *pvl* gene, PCR was performed. For the antibiotic resistance genes: *blaZ* (penicillin), *mecA* (oxacillin) and *vanA* (vancomycin) were targeted (Zehra et al. 2017, 2019a; Figures S1–S3,

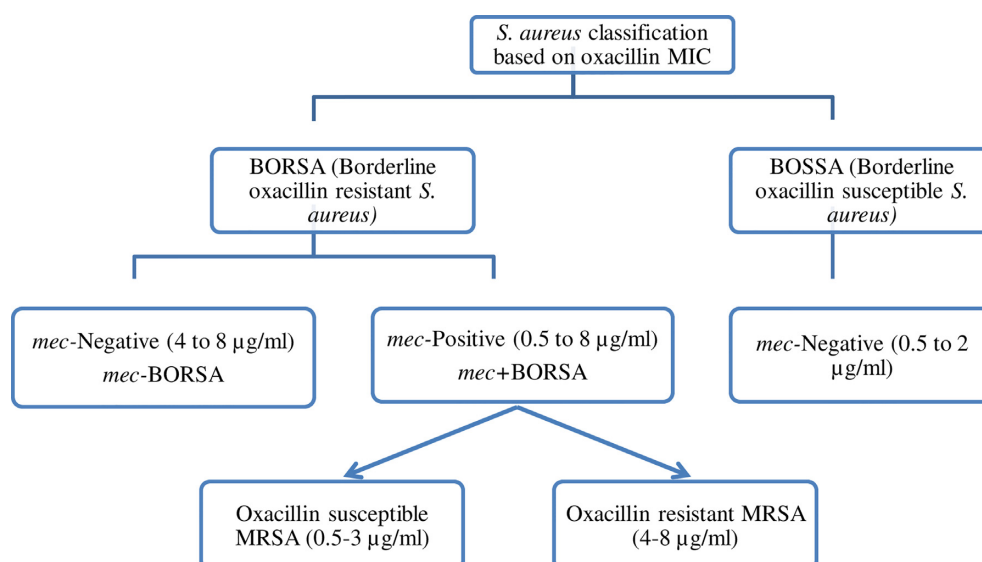


Figure 1. Flow diagram showing the classification of *S. aureus* based on the *mecA* gene and oxacillin MIC.

Table 1. Prevalence of *S. aureus* in meat, milk and community settings from Punjab, India.

S. no.	Sample type	<i>S. aureus</i> n/N (%)	MRSA (%)	Total isolates	Selected isolates
1.	Chicken	47/172 (27.3)	5/172 (2.9)	191	54
2.	Chevon	27/148 (18.2)	0	114	40
3.	Pork	30/147 (20.4)	0	127	47
4.	Milk	39/214 (18.22)	7/214 (3.3)	126	62
5.	Community	59/269 (21.9%)	20/269 (7.4)	203	74
6.	Total	202/950	32/950 (3.4)	761	277

Supporting Information). Primers used in this study are mentioned in Table S3 (Supporting Information).

Staphylococcus aureus isolates ATCC 33591 (*mecA* and *pvl* + ve) and ATCC 33592 (*mecA* and *pvl* -ve) were used as MRSA and *pvl* control. Standard strain ATCC25923 was used as MSSA (*mecA* -ve) control. Strain ATCC 700699 was used as a control for SCC*mec* type II and reduced vancomycin susceptibility. *S. aureus* field isolates (Accession nos. KU872013, KP896298, KP610412 and KT005393) from a previous study (Zehra et al. 2019a) were used as a positive control for genes *blaZ*, *pvl*, SCC*mec* type IVa and SCC*mec* type V, respectively.

Statistical analysis

Statistical analysis was conducted using IBM SPSS Statistics v.22.0 (IBM Corp., Armonk, NY). For analysis of categorical data, isolates showing intermediate resistance were classified as 'resistant/non-susceptible'. The logistic regression model using the antibiotic resistance status of each isolate as a binary outcome i.e. susceptible and non-susceptible (combining intermediate resistance (IR) and resistance (R)) or gene present and absent with sample type as explanatory variables, was used to model resistance profiles modulated by the type of sample. For each antibiotic, the odds of the isolate being non-susceptible or positive for the gene under study were computed against sample type.

All the statistical analyses were performed at a significant value of $P < 0.05$. The χ^2 test was performed to describe the associations between an individual *in silico* detection of acquired

pvl, antimicrobial resistance genes and ceftriaxone, cotrimoxazole, vancomycin MIC levels at its respective non-susceptible threshold. The Fisher-Freeman-Halton exact test of independence was performed and the phi (ϕ) or Cramer's V (ϕ_c) coefficient was calculated to describe the correlations. Cochran's-Mantel-Haenszel statistics was performed to estimate the association between antimicrobial resistance genes and phenotypic resistance after adjusting for the type of sample.

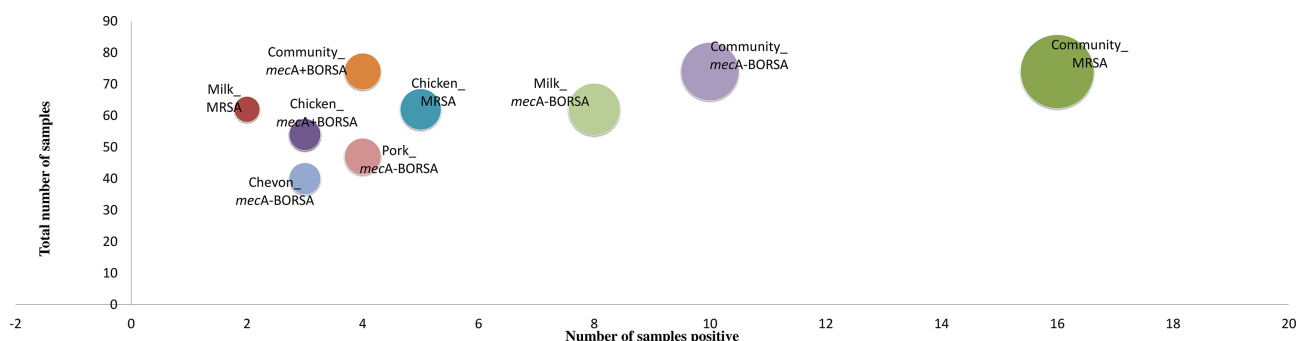
Sensitivity analyses were implemented to comprehend the impact of classifying intermediate isolates as susceptible instead of resistant. Logistic regression models were utilized to evaluate the odds of resistance prevalence in community settings when contrasted with the food of animal origin for every antimicrobial. In this study, no significant changes were observed in the direction and statistical significance of associations between sample type and patterns of antimicrobial resistance or between the sample type and gene presence/absence.

RESULTS

The prevalence of *S. aureus* in meat and milk samples has already been published in our previous studies (Kaur et al. 2019; Zehra et al. 2019a,b). Although, the overall prevalence of *S. aureus* (MRSA) in the food of animal origin was 21% (95% confidence interval (CI) 17.9–21.4%; 1.8% MRSA) that accounts for 22.3% (95% CI 18.5–26.1%) and 18.2% (95% CI 13–23.4%) in meat and milk samples, respectively (Table S2, Supporting Information). In community settings, the prevalence was 21.9% (95% CI

Table 2. Frequency distribution of MRSA and BORSA isolates.

Source	No. of isolates	Based on <i>mecA</i> and oxacillin (MIC $\mu\text{g/mL}$)			
		<i>mecA</i> ⁺ MRSA (%)	<i>mecA</i> ⁺ BORSA (%)	<i>mecA</i> ⁻ MSSA (%)	<i>mecA</i> ⁻ BORSA (%)
Chicken	54	3 (5.5)	3 (5.5)	48 (88.89)	0
Chevon	40	0	0	37 (92.5)	3 (7.5)
Pork	47	0	0	43 (91.49)	4 (8.51)
Milk	62	2 (3.2)	5 (8.06)	47 (75.81)	8 (12.9)
Community	74	16 (21.6)	4 (5.41)	44 (59.46)	10 (13.51)
Total	277	21 (7.58)	12 (4.33)	219 (79.06)	25 (9.02)

**Figure 2.** Distribution of MRSA and BORSA isolates from the meat, milk and community samples, disaggregated by frequency of *S. aureus* positive samples and total sample size of each sample type.

17–26.8%; 7.4% MRSA; Table 1). District-wise distribution of the *S. aureus* is given in Table S2 (Supporting Information).

Among the selected 277 isolates, six from meat samples, seven from milk and 20 from the community were MRSA (*mecA* positive). The oxacillin MIC estimations of MRSA isolates ranged from 0.75 to 32 $\mu\text{g/mL}$ and there was significant variation in oxacillin MIC values that were higher in community settings than in the food of animal origin ($P < 0.05$; Figure S4, Supporting Information). In this study, out of these 33 MRSA isolates, 12 (three from meat, five from milk and four from the community) demonstrated MIC extend from 0.75 to 8 $\mu\text{g/mL}$ thereby representing *mecA*⁺BORSA (Fig. 1). MRSA isolates from meat were SCCmec type V and *pvl* positive (SCCmecV *pvl*⁺). From milk, four isolates were SCCmecV *pvl*⁺ and three were non-typeable (NT) *pvl*⁺. Varying nature of MRSA was found among community isolates ($n = 20$), where the 10 isolates were SCCmec IVa, among which nine were *pvl*⁺; the rest of the 10 isolates were NT by SCCmec typing, yet eight isolates were *pvl*⁺.

Isolates from the community setting were almost three times more inclined to be MRSA than the isolates from chicken (OR = 2.963, 95% CI 1.099–7.988) and milk samples (OR = 2.910, 95% CI 1.138–7.443; Table 2 and Fig. 1). This odds value for community isolates was 5.33-fold (95% CI 2.529–11.585) greater when contrasted with food of animal origin. None of the isolates from pork and chevon was positive for the *mecA* gene and there was no statistically significant difference among isolates from meat and milk samples in terms of *mecA* gene ($P > 0.05$). Similarly, there was statistically no significant difference between *pvl* positive isolates from meat and milk aside from the community isolates. The likelihood of discovering *pvl*⁺MRSA isolates was 4-fold higher in the community setting than in the food of animal origin (95% CI 1.997–9.513; Fig. 2). Additionally, there was a positive correlation between *pvl* and *mecA* gene ($\phi = 0.706$, $P = 0.000$).

An aggregate of 25 isolates was *mecA*⁻ BORSA with MIC ranging from 4 to 8 $\mu\text{g/mL}$ aside from the two isolates from a community (mobile phone) that demonstrated MIC = 12 $\mu\text{g/mL}$ (Fig. 1 and Table 2). Such example might be a result of *mecA* variant genes like *mecB*, *mecC* or hyperproduction of β -lactamases, production of typical PBP (penicillin-binding protein) with altered binding capacity (Becker et al. 2018; Cikman et al. 2019). To reject the likelihood of hyperproduction of β -lactamase, co-amoxiclav antibiotic disk diffusion test was performed. The results showed that isolates from chevon (oxacillin resistant, *blaZ* positive, *mecA* negative) were ceftriaxone susceptible and β -lactamases hyper-producers with inhibition zone more than 20 mm yet uneven outcomes were found if the isolates were from pork and community settings.

The likelihood of finding beta-lactamase hyper-producers was observed to be 2-fold higher in community settings than in meat sample (OR = 2.991, 95% CI 1.089–8.218, $P = 0.032$) however this distinction was non-significant when contrasted with isolates from milk samples ($P > 0.05$; Table 2 and Fig. 3). These isolates were significantly correlated with the ceftriaxone MIC values ($\phi = 0.277$, $P = 0.000$). The ceftriaxone MIC values were lower in the beta-lactamase hyper-producers than non-beta-lactamase hyper-producers (OR = 1.034, 95% CI 1.008–1.060, $P = 0.011$).

Also, no significant correlation was found between isolates harboring *pvl* gene and *mec*⁻ BORSA. Albeit, significant correlation was observed between isolates carrying *pvl* and *mec*⁺ BORSA ($\phi = 0.413$, $P = 0.000$). Similarly, for co-detection of *pvl* and cotrimoxazol resistance, a moderately strong yet significant correlation was found between *pvl*⁺ isolates and their cotrimoxazol resistance ($\phi = 0.547$, $P = 0.000$) and the chances of cotrimoxazol resistance were 14.32-folds higher among the *pvl* carrying isolates (95% CI 6.827–25.03, $P = 0.000$).

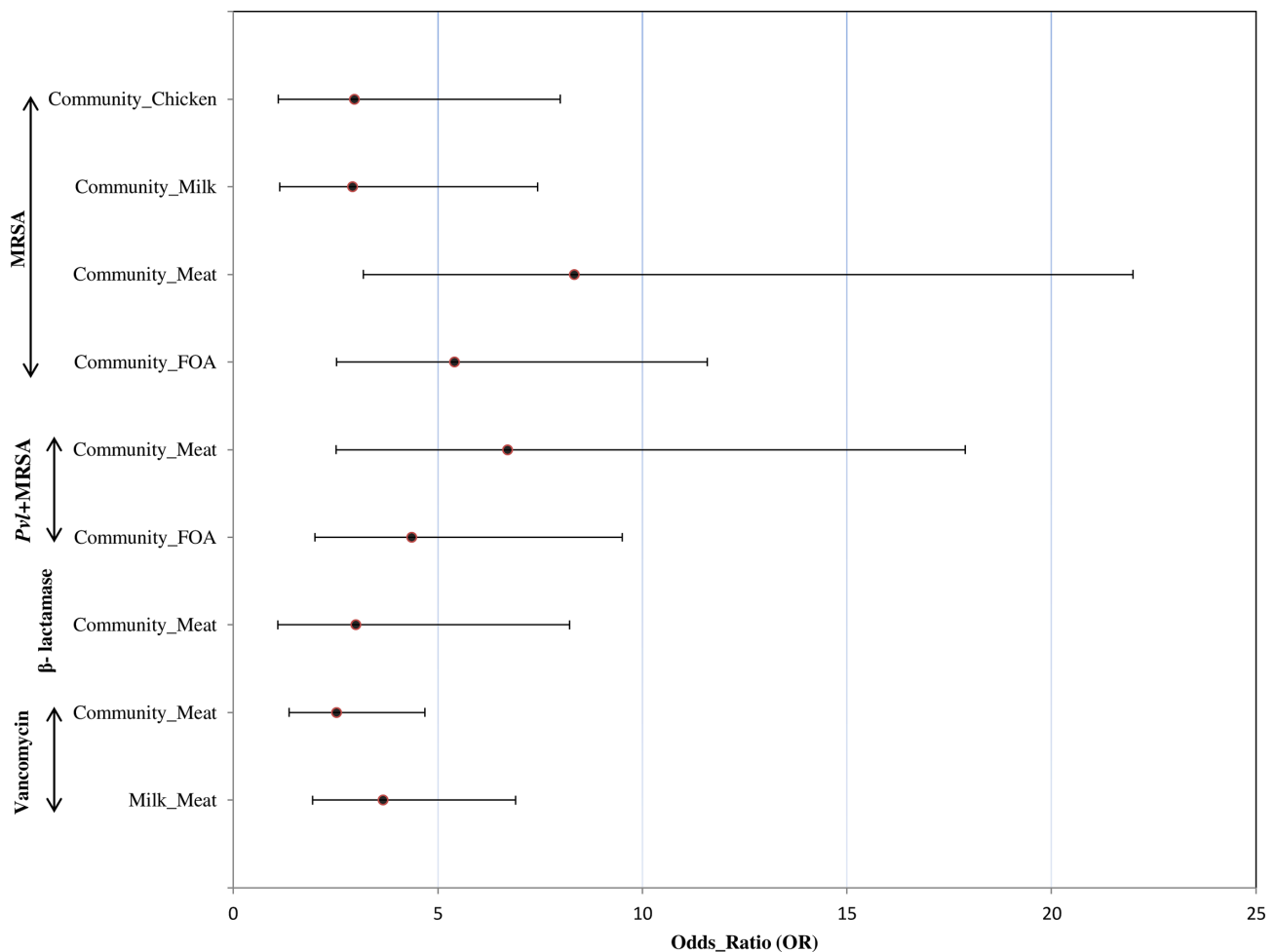


Figure 3. According to the outcomes from logistic regression modelling the risk of vancomycin, β -lactamase hyperproducers, *pvl*⁺MRSA and MRSA prevalence against sample type. The figure displays the risk (in terms of odds ratios) of MRSA, *pvl*⁺MRSA and β -lactamase hyperproducers in community settings against the chicken, meat, milk and food of animal origin (FOA). For the analyses, intermediate resistant isolates were treated as resistant. The horizontal lines represent 95% confidence intervals for the estimated odds ratios. The 95% CI for the OR comparing community versus meat is wide (3.174–21.881). This is because of the way that there were few MRSA in meat samples (only chicken samples were certain for MRSA). While the OR is statistically significant, the CI suggests that the magnitude of the impact could be somewhere in the range of 3-fold higher to 22-fold higher. A large study is needed to produce a more precise estimate of impact.

Lastly, none of the isolates was resistant to vancomycin neither had *vanA* gene that encodes for such resistance. Although, there was a significantly frail correlation between the MRSA and vancomycin MIC values ($\phi = 0.286$, $P = 0.002$). The correlation of vancomycin MIC was strong with the type of sample ($\phi = 0.651$, $P = 0.000$). Higher vancomycin MIC (2–4 $\mu\text{g/mL}$) was seen among the isolates from milk and community. The higher vancomycin MIC was 2.52-folds more probable in community isolates ($\text{MIC} \leq 4 \mu\text{g/mL}$) and was 3.66-folds more probable in milk isolates ($\text{MIC} \leq 2 \mu\text{g/mL}$) compared to isolates from meat samples (Fig. 3).

DISCUSSION

Overall, 22.3% meat, 18.2% milk and 21.9% community samples were positive for *S. aureus*. Over the 14 districts of Punjab, *S. aureus* was confirmed ranging from 16.9% in Barnala to 25% in Sangrur, irrespective of sample type. This suggests the ubiquitous distribution and high prevalence of *S. aureus* in Punjab.

MRSA has been isolated from retail milk, meat and milk/meat industry workers and has also been reported from community settings around the globe. The same has been

observed in this study, where the MRSA was found in chicken, milk and community settings. SCCmec type V was predominant among the food of animal origin while SCCmec type IVa was common among the community settings. These outcomes are in concordance with other studies (Yang et al. 2016; Giovanni et al. 2020) and exhibit significant public health concern in light of the fact that SCCmec type IV and type V are found as predominant CA-MRSA in the hospital settings of India (Rajan et al. 2015; Singh et al. 2017). Not only that, but this study also reports the high prevalence of *pvl* positive MRSA in the food of animal origin that is in-contrary to the other studies (Sato et al. 2017; Giovanni et al. 2020) but in concordance with a study like Wu et al. 2019. This is quite worrisome for the reason that PVL is known to be associated with more severe infections (Gijón et al. 2016).

This study also reported the *mecA*⁺BORSA that were mainly beta-lactamase hyper-producers except few from pork samples that were *mecA*⁺BORSA yet not beta-lactamase hyper-producers; although not significant. These isolates from pork were intermediate or completely resistant to ceftriaxone. This can be clarified by another explanation behind being *mecA*⁺BORSA that is modified *S. aureus* (MODSA) with mutations in *pbp* genes and/or PBP4-overexpression (Argudín et al. 2018) and need to be further

analyzed for the presence of *mecA* variants or any mutation in *pbp* genes which was limited in this study.

According to the literature, unlike CA-MRSA isolates, BORSA strains lack PVL locus responsible for the expression of the Panton-Valentine Leukocidin (Hryniewicz and Garbacz 2017). However, in this study, no such relationship was found and BORSA isolates with or without the *pvl* gene were observed. This may be because the BORSA isolates in this study were *mecA*^{+/−} (MIC 0.5–8 µg/mL) however in the previous studies BORSA isolates were *mecA*[−] only (MIC 0.5–8 µg/mL). Although, it was observed in the present study that the occurrence of the *pvl* gene in methicillin-susceptible *S. aureus* (MSSA) was comparable to the occurrence of the *pvl* gene in BORSA. Also, co-detection of *pvl* and cotrimoxazol resistance was reported in this study which is in concordance with the published study (Kraef et al. 2015). There was, however, no noteworthy correlation of cotrimoxazol resistance with the type of sample but the positive correlation between the *pvl* positive *S. aureus* and cotrimoxazol-resistance is of concern because cotrimoxazol prophylaxis may put patients (particularly HIV patients) at the danger of SSTIs.

Vancomycin remains the antibiotic of choice to treat numerous MRSA infections. However, because of the striking ascent in MRSA infections and broad utilization of vancomycin, MRSA strains with reduced susceptibility to vancomycin are rising. In this study, none of the isolates was resistant to vancomycin neither had *vanA* gene that encodes for such resistance but the increase in the vancomycin MIC among the MRSA isolates was observed (Vancomycin MIC creep). This is problematic because the patients infected with MRSA isolates that display MIC creep may encounter less fortunate clinical results, including delayed response to treatment, high mortality, protracted hospitalization or by and large expanded expense of hospitalization (Diaz et al. 2018; Kok et al. 2018). This study also recorded the vancomycin creep among MSSA isolates. Kok et al. also reported that the vancomycin MICs of ≥2 µg/mL were related to unfavorable clinical results in MSSA independent of anti-microbial choice, recommending this might be an alternate for organism virulence.

In conclusion, to the best of our knowledge, this study is only among the few that report the variation in the occurrence of MRSA and BORSA from a different type of samples. This study also documented the high prevalence of *pvl* positive SCCmec V/IV MRSA in meat, milk and community settings in Punjab, India. Likewise, this study also reported the prevalence of BORSA classified as *mecA*⁺BORSA and *mecA*[−]BORSA. No in-depth study concerning the prevalence of BORSA strains and its association with sample type has been published up to this point. Additionally, a positive relationship between the *pvl*⁺ *S. aureus* and cotrimoxazol/vancomycin MIC is of a concern because the patients infected with such isolates may encounter less fortunate clinical results.

AUTHOR'S CONTRIBUTIONS

Zehra A.– Lab work, data analysis and manuscript writing, Gulzar M.– sampling and SCCmec typing of isolates, Singh R. and Kaur S.– a technical program of work.

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SUPPLEMENTARY DATA

Supplementary data are available at [FEMSLE](https://femsle.onlinelibrary.wiley.com/doi/10.1111/femsle.13711) online.

Conflicts of Interest. None declared.

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